

STAT 452/652 Predictive Modeling Project Report

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Overview

This project builds and compares predictive models for a continuous response Y using 20,000 training observations and 19 numeric predictors. The objective was to minimize mean squared prediction error on a separate test set.

Methods Evaluated

- Ordinary least squares
- Ridge regression
- LASSO regression
- Elastic net regression
- Principal components regression
- Partial least squares regression
- Regression trees
- Random forest
- Gradient boosting machines

Model Comparison

Model	Performance	Notes
OLS	~5.19 RMSE	Baseline
Ridge	~5.22 RMSE	Slightly worse
LASSO	~5.19 RMSE	Similar to OLS
Elastic Net	~26 MSE	Not competitive
PCR	~5.19 RMSE	No major gain
PLS	~5.19 RMSE	Similar
Regression Tree	~26 MSE	Underfit
Random Forest (CV)	4.01 RMSE	Strong performer
Tuned Random Forest (OOB)	15.84 MSE (~3.98 RMSE)	Best overall
GBM	4.44 RMSE	Good but weaker

Final Model

The best model was a manually tuned random forest with $mtry = 11$, $nodesize = 2$, and $ntree = 1500$ in the final fit. It achieved the lowest estimated prediction error among the candidate models and was used to generate the final test-set predictions.

Important Predictors

Random forest importance and LASSO coefficient behavior suggested that the most influential predictors included X11, X13, X1, X8, X12, X5, X15, X9, and X19. Combining both sources of evidence, roughly 8 to 10 predictors appeared meaningfully influential.

Conclusion

The modeling workflow shows that the relationship is better captured by nonlinear ensemble methods than by standard linear, shrinkage, or dimension-reduction approaches. The tuned random forest provided the best balance of predictive accuracy, stability, and reproducibility for this dataset.